# Usability Testing

| |
|---|
| *"You've stumbled across an article about biodiversity today: give it a read."*<br>*"Tell me what kind of picture you think it paints of humanity and biodiversity."*<br>**Objectives**<br>• Determine if users take away the intended point, i.e. humanity is driving mass biodiversity loss |
| *"You're only interested in this article's graphs. Switch between the charts quickly and try to vocalise your reactions to navigating and also to each chart, particularly what you think each is about."*<br>**Objectives**<br>• Determine if the charts are comprehensible, intuitive and scannable |

I gave both tasks to **8 participants** via a mix of zoom and in-person. I tried to vary the participants by how much each knew about human-induced biodiversity loss, so some would be coming in fresh, and others would know the facts but would still be (hopefully) bemused by the presentation regardless.

## Results & changes

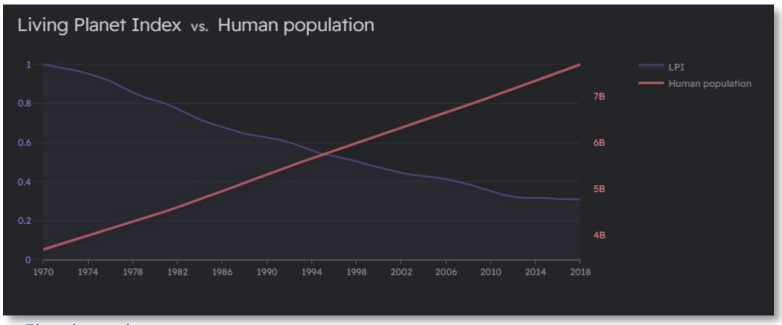
**Task 1** was a success. Users appeared to mirror the point of the article quite well, vocalising complex feelings about humanity's influence, and sadness — one user even stated they felt 'regret' — over the loss of biodiversity. No changes were made from this feedback.

**Task 2**, on the other hand, prompted a wealth of changes to the charts throughout testing, as I continually edited based on initial feedback, and tested new changes with the following participant.

### Chart 1

I changed chart 1 thrice across the eight tests:

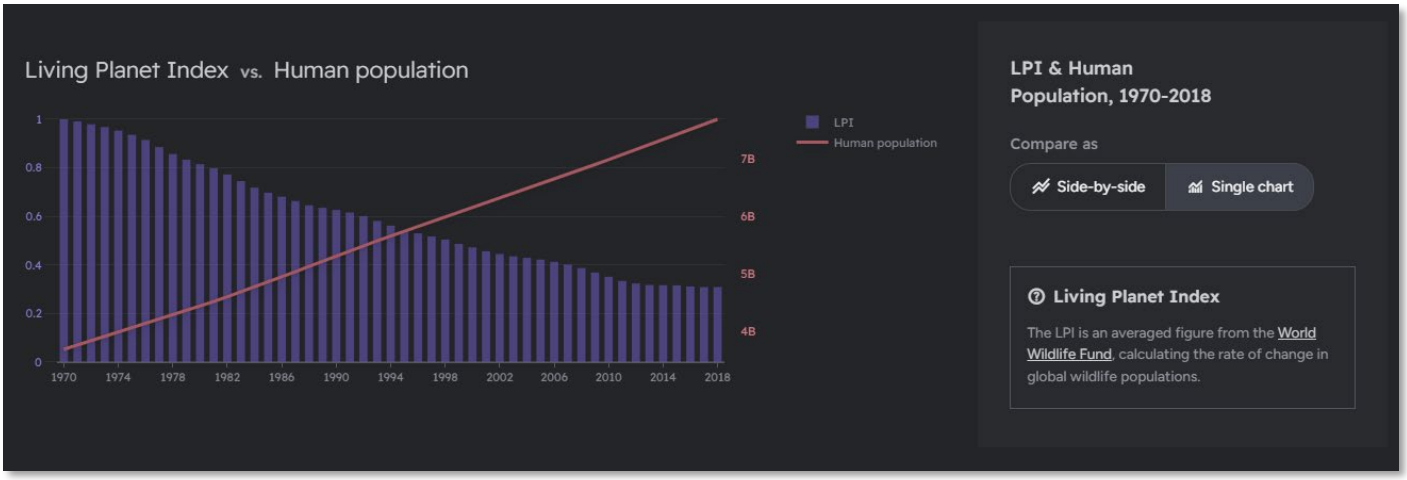
- **The first iteration** was a dual-Y axis line chart, contrasting the two lines. However, an initial user indicated confusion at first glance of the chart — I had been harbouring doubts about the dual-Y format myself.
- **So, I changed the first chart to two side-by-side charts**, showing the different data points. Although this appeared not to cause the same confusion, users reported that the comparison was not as immediately digestible as it was in the first iteration.
- **I decided the best way forward** was to combine the two above and give users the choice of charts.



*First iteration*

*Second iteration*

    - This decision elicited positive reactions from the final two users, who appreciated the depth of interaction and choice.




*Final iteration control panel and dual axis*

### Chart 2

A chart with complicated interactions, it received two iterations:

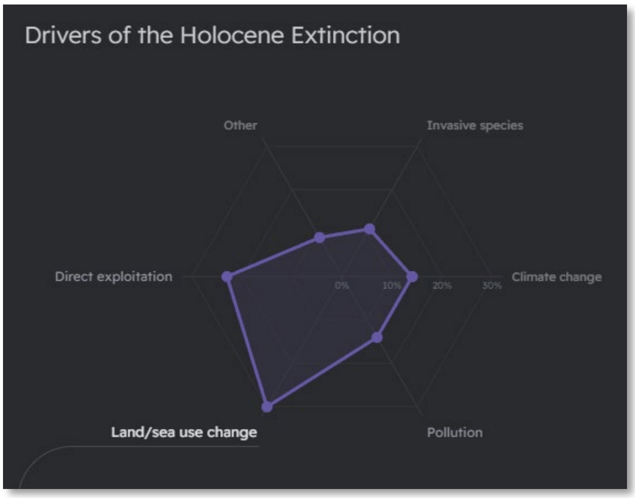
- **The first version** had centred header text and play buttons. Users did not instantly use the slider and expressed minor irritation at the readability of this version.
    - I also noticed that users avoided the play/pause buttons, and when asked, said they felt no need to use them.
- **The final version** changed the text to better fit the flow, removed the buttons, and added a tooltip to highlight the slider.


*First iteration*

### Chart 3

A simple chart, but a non-standard one. I updated it only once:

- **Early users reported** that they felt confused at the message of the chart, particularly how it was supposed to single out humans.
- **So, the final iteration** split the centre radar into two groups, which seemed to solve the issue.




*First iteration*

# Visualisation Narrative

Biodiversity, the concept of biological variance in flora/fauna, is the bedrock on which all life on Earth evolved and flourished — the article attempts to convey that this bedrock is being eroded, by humanity itself. Since my article tackled the murder of biodiversity, with humanity as prime suspect, I decided that the structure of the article should be roughly that of a judicial case. Detailing first humanity's capacity and record; then elaborating on the means by which biodiversity is suffering — this would make my argument more convincing by establishing a logical flow for the article. Hopefully, this would be enough to convince readers unaware of the issues, while evoking the intended emotional reaction of horror and cynicism.

To achieve this logical flow, I chose data in an iterative fashion, meaning each new datapoint would complement and build on the last one in some way:

First, I thought an interesting hook would be a line chart — a great way to represent temporal increases/decreases in data — contrasting human population (Roser et al., 2013) and general biodiversity health (WWF, 2022). Very little data wrangling was required since the datasets were CSVs that followed similar timespans, although both had a few irrelevant removed.

Then, to further establish background, I chose a contrast between human migration and major extinctions (Andermann et al., 2020) — a choropleth was ideal for this worldwide data. A large amount of data wrangling was required here. The original data was scattered across tables, so it had to be normalised, hand-typed, and mapped across spreadsheets.

Next, I wanted to lead more into the present, and the idea of the Holocene Extinction. To introduce this and elaborate from the first two on *how* humans affect biodiversity, I sought out data on the current drivers of biodiversity loss (IPBES, 2019). I thought a radar chart would be the ideal solution, as it enabled non-temporal comparison between A. the different drivers and B. the share of human vs natural driver responsibility. A radar also allowed me to both visually distinguish the most important variable — *land/sea use change* — and separate it from the other elements, in order to attach guiding lines from it to the next chart.

Land/sea use change was humanity's main driving force behind their destruction of biodiversity, so I wished to elaborate upon it, particularly pointing out the fact that humanity is expanding and not slowing down. So, a 100% stacked area chart was chosen. This would succinctly contrast human-influenced land versus natural territory, while also being able to chart the temporal growth of humanity's share.

And finally, I wanted to end the article on a note on the future. In keeping with the tone thus far, the last chart would be a series of grim predictions to help the article's point linger in the mind of readers. Biodiversity losses from climate change (Strona & Bradshaw, 2022) were the perfect way to do this, showing the end result of current human activity. Luckily, the raw data was easily accessible, and little wrangling was needed.

# References


## Data

Roser, M., Ritchie, R., Ortiz-Ospina, E., & Rodés-Guirao, L. (2013). World Population Growth. *Our World In Data*. ourworldindata.org/world-population-growth

World Wildlife Fund (WWF), Zoological Society of London (ZSL). (2022). *The Living Planet Report*. www.livingplanetindex.org/latest_results

Andermann, T., Faurby, S., Turvey, S. T., Antonelli, A., & Silvestro, D. (2020). The past and future human impact on mammalian diversity. *Science Advances, 6*(36). doi.org/10.1126/sciadv.abb2313

Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services (IPBES). (2019). *Global assessment report on biodiversity and ecosystem services*. Brondizio, E. S., Settele, J., Díaz, S., Ngo, H. T. (eds.). IPBES secretariat, Bonn, Germany. doi.org/10.5281/zenodo.3831673

Ellis, E.C., Beusen, A.H., & Goldewijk, K.K. (2020). Anthropogenic Biomes: 10,000 BCE to 2015 CE. *Land (Basel), 9*(5), 129–. doi.org/10.3390/land9050129

Strona, G. & Bradshaw, C.J.A. (2022). Coextinctions dominate future vertebrate losses from climate and land use change. *Science Advances, 8*(50). doi.org/10.1126/sciadv.abn4345

*The below are referenced in index.html:*

## Images

Bishop, J. (2018). *Sun light passing through green leafed tree* [Photograph]. unsplash.com/photos/098SVhpuA5Q

Shwarz, A. (2018). *Time-lapse photography of city night lights* [Photograph]. unsplash.com/photos/XS7q-baZrmE

## Inspirations

Ritchie, H. (2022). Did humans cause the Quaternary Megafauna Extinction. *Our World In Data*. ourworldindata.org/quaternary-megafauna-extinction

## Other

hrbrmstr. (2015). Continents GeoJSON [Github dataset]. https://gist.github.com/hrbrmstr/91ea5cc9474286c72838